

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		37 <u>years</u> Don't accept 37 light years Accept 1 169 200 000 <u>s</u> or 13 505 <u>days</u>	1			1		
		(ii)		Substitution: $3 \times 10^8 = \frac{d}{31600\,000 \times 37}$ ecf (1) Rearrangement: $d = 3 \times 10^8 \times 31\,600\,000 \times 37$ (1) $= 3.5 \times 10^{17}$ [m] (1) Award 2 marks for an answer of 9.48×10^{15} [m] Award 2 marks for an answer of 1.11×10^{10} [m]	1	1 1		3	3	
	(b)	(i)		Any 2 × (1) from: - Arcturus is <u>bigger</u> or has a <u>bigger radius</u> - Arcturus is <u>brighter</u> or <u>more luminous</u> - Arcturus is <u>cooler</u> or has a <u>lower temperature</u> Accept the converse answers. Don't accept reference to 'it'.		2		2		
		(ii)		Hydrogen reduces / hydrogen runs out / hydrogen fusion stops (1) Helium fusion begins (1) Increasing gas or radiation pressure (1) As outwards force is larger [than the inwards force] (1) Don't accept Sun has burnt all of its hydrogen	4			4		
				Question 5 total	6	4	0	10	3	0